REVISION NOTES

Week 5

Sun 15 – Fri 20 November 2026

14	10 h 12 min	13 / 14
LECTURES	VIDEO	SUMMARISED

COURSES THIS WEEK

MATH 120	Single-Variable Calculus	2 lectures · 1 h 37 min
AI 200	Neural Networks: Zero to Hero	2 lectures · 4 h 13 min
SPCH 100	Storytelling	6 lectures · 1 h 13 min
MATH 130	Probability	2 lectures · 1 h 40 min
MATH 110	Linear Algebra	2 lectures · 1 h 30 min
Total		**14 lectures · 10 h 12 min**

Study days Sun 15 Nov to Fri 20 Nov. Hours are video time; a lecture the plan spreads over several days counts only the days that fall in this week.

WEEK 5 · SUN 15 – FRI 20 NOVEMBER 2026

Contents

LECTURES, IN PLAN ORDER

1	MATH 120	Implicit Differentiation		Sun 15
2	AI 200	Let's reproduce GPT-2 (124M)		Sun 15 – Wed 18
3	SPCH 100	Personal signature stories — make them, then bank them		Sun 15
4	MATH 130	Expectation, Indicator Random Variables, Linearity		Mon 16
5	SPCH 100	Twelve lessons: zoom in, quote it, use 'how are you?'		Mon 16
6	SPCH 100	Most of the work is finding — and killing		Mon 16
7	MATH 110	Solving Ax = 0: Pivot Variables, Special Solutions		Tue 17
8	SPCH 100	The taste gap — volume on a deadline		Tue 17
9	SPCH 100	The voice toolbox — register, prosody, pace, and a warm-up		Tue 17
10	MATH 120	Exponential and Log		Wed 18
11	SPCH 100	The five dials — rate, volume, melody, tone, pause		Wed 18 – Thu 19
12	MATH 130	Expectation Continued		Thu 19
13	AI 200	Intro to Large Language Models	No summary	Thu 19 – Fri 20
14	MATH 110	Solving Ax = b: Row Reduced Form R		Fri 20

AT THE BACK

Not yet summarised	1 lecture
What this week established	
Answers to the self-checks	

13 of 14 lectures have a summary. The other 1 are listed, with their video links, on “Not yet summarised”; nothing is written for them here.

MATH 120 · SINGLE-VARIABLE CALCULUS · LECTURE 1 OF 14

Implicit Differentiation

Sun 15 Nov · 49 min video · Watch on YouTube

THE TAKEAWAY

Differentiate the equation you were given instead of solving it first, and both fractional powers and every inverse function fall out.

THE ARGUMENT · 6 STEPS

1. **Fractional powers, via an integer equation**
   $y = x^{m/n}$ is not yet differentiable by anything you know. But $y^n = x^m$ is. Differentiate that with the chain rule, solve for y', substitute y back — and the exponent arithmetic collapses to $\frac{m}{n}x^{m/n-1}$. The power rule now covers every rational exponent.

2. **The method in one line**
   Apply $\frac{d}{dx}$ to BOTH sides of the equation, treating y as a function of x so every y term picks up a y'. Then solve for y' algebraically.

3. **On the circle it beats the explicit route**
   $x^2 + y^2 = 1$ differentiates to $2x + 2yy' = 0$, so $y' = -x/y$ — three symbols. Solving for y first forces a square root, a chain rule, and a choice of branch, and the answers agree.

4. **And it handles both branches at once**
   $-x/y$ is correct above and below the axis without comment: on the lower half y is negative, so the slope comes out positive, exactly as the picture shows.

5. **The gain grows with the mess**
   $y^4 + xy^2 - 2 = 0$ has a quartic solution with nested radicals and up to four branches. Implicitly it is two lines. You still need a point — at $(1, 1)$ the slope is $-\frac{1}{6}$ — but you never have to DIFFERENTIATE the monstrous formula.

6. **Every inverse function, for free**
   $y = \arctan x$ means $\tan y = x$. Differentiate: $\frac{y'}{\cos^2 y} = 1$, so $y' = \cos^2 y$. Draw the right triangle with legs x and 1, read $\cos y = 1/\sqrt{1+x^2}$, and the answer is $\dfrac{1}{1+x^2}$.

WORKED EXAMPLE

The same three steps give $\frac{d}{dx}\arcsin x$. From $\sin y = x$: $(\cos y)y' = 1$, so $y' = 1/\cos y$, and the triangle gives $\cos y = \sqrt{1-x^2}$. Answer $\dfrac{1}{\sqrt{1-x^2}}$.

WATCH FOR

Leaving the answer in terms of y. $\cos^2(\arctan x)$ is correct and useless; the triangle is what converts it into a formula in x alone, and for trig functions it always simplifies.

CHECK YOURSELF · ANSWERS AT THE BACK

1.1 Implicit differentiation is preferred for $y^4 + xy^2 - 2 = 0$ because:

A The explicit solution does not exist
B You never have to differentiate the quartic formula — two lines give y' in terms of x and y
C It avoids the chain rule
D It gives a simpler curve

1.2 The derivative of $\arctan x$ at $x = 2$ is:

Your answer: _____

AI 200 · NEURAL NETWORKS: ZERO TO HERO · LECTURE 2 OF 14

Let's reproduce GPT-2 (124M)

Sun 15 Nov – Wed 18 Nov · day 2–5 of 5 · 241 min video · Watch on YouTube · began in week 4

THE TAKEAWAY

Reproducing GPT-2 (124M) is the GPT from before plus everything that makes it real: the exact architecture so OpenAI's weights load, GPT-2's initialisation, a long list of speed-ups, GPT-3's optimisation recipe — AdamW, clipping, warmup and cosine decay, a 0.5M-token batch via gradient accumulation — and honest evaluation on held-out loss and HellaSwag.

THE ARGUMENT · 8 STEPS

1. **Match GPT-2 exactly, then load it**
   12 layers, 12 heads, n_embd 768, context 1,024, vocabulary 50,257 (50,000 merges + 256 bytes + end-of-text). Module names follow Hugging Face's, so from_pretrained copies OpenAI's weights in — transposing the Conv1D ones — and the model samples sensible text before any training.

2. **Data as shifted token buffers**
   A buffer of B·T + 1 tokens becomes x (all but the last) and y (all but the first), viewed as (B, T). The loss is cross-entropy over every position. A tiny batch can be overfitted to check the pipeline before training for real.

3. **Weight tying and GPT-2's init**
   The token embedding wte and the output lm_head share one matrix — GPT-2's checkpoint already does. Weights start at std 0.02, and the projections that write into the residual stream are scaled by $1/\sqrt{2 \cdot n_{\text{layer}}}$, because each block adds two contributions and the stream's std would otherwise grow with depth.

4 **Make it fast**

TF32 matmuls, bfloat16 autocast, torch.compile, flash attention through F.scaled_dot_product_attention, and the vocabulary padded from 50,257 to the 'nice' 50,304 — each its own commit, each measured in tokens per second.

5 **GPT-3's optimisation recipe**

AdamW with betas (0.9, 0.95) and eps 1e-8, gradient norm clipped at 1.0, a learning rate that warms up linearly for 715 steps to 6e-4 and then decays on a cosine to 6e-5, and weight decay 0.1 on 2-D parameters only.

6 **A 0.5M-token batch on any hardware**

total_batch_size is 524,288 tokens; with micro-batches of B·T tokens the gradient is accumulated over several forward/backward passes before one optimizer step, and the loss is divided by the number of micro-steps so the accumulated gradient is a mean, not a sum. DDP splits the micro-steps across GPUs and averages.

7 **Real data: FineWeb-Edu**

The 10B-token sample, tokenized with GPT-2's tokenizer, end-of-text between documents, stored as uint16 in 100 shards of 100M tokens, the first shard kept for validation. 19,073 steps of 0.5M tokens is about one epoch.

8 **Evaluate honestly**

Validation loss every 250 steps, and HellaSwag in completion style: each of the four endings is appended to the context and the one with the lowest average per-token loss is chosen. The 124M run is compared with OpenAI's GPT-2 checkpoint (val 3.2924; HellaSwag 0.2945) and GPT-3 124M (0.337).

WORKED EXAMPLE

When changing the batch size, keep the objective the same: fix total_batch_size in tokens, derive grad_accum_steps = total / (B · T · world_size), and divide each micro-step's loss by grad_accum_steps. Then the gradient matches the big batch's mean whatever micro-batch fits in memory — the play notebook checks exactly this on a tiny MLP.

WATCH FOR

Accumulating gradients without rescaling the loss. backward() adds gradients, so k micro-steps produce the gradient of a SUM of k mean losses — k times too large — which silently multiplies the effective learning rate by k.

CHECK YOURSELF · ANSWERS AT THE BACK

2.1 GPT-2 (124M) ties its token embedding to its output layer: vocab 50,257, n_embd 768. How many parameters does sharing that one matrix save? (digits only, no commas)

Your answer: _____

2.2 _init_weights multiplies the std of every c_proj layer by (2 · n_layer)^−0.5. Why?

A To make those projection layers learn faster than the rest of the network
B Each block adds two outputs to the residual stream; this stops its std growing
C To compensate for the GELU nonlinearity, which halves the variance of its input
D Because the projection layers have no bias, so they need less noise

2.3 total_batch_size = 524,288 tokens, micro-batch B = 64 sequences of T = 1,024 tokens, on one GPU. How many gradient-accumulation micro-steps make one optimizer step?

Your answer: _____

SPCH 100 · STORYTELLING · LECTURE 3 OF 14

Personal signature stories — make them, then bank them

Sun 15 Nov · 13 min video · Watch on YouTube

THE TAKEAWAY

Aaker's family rules for creating signature stories, not just finding them: carve out areas of incompetence, be sneaky and brand ordinary days as special, seek highs and lows, and bank the results — six words is enough.

THE ARGUMENT · 5 STEPS

1. **Carve out areas of incompetence**
   Her father cannot take a photograph; her husband needs a full protocol before he can take the bins out. Cheerful incompetence generates stories — and spares you the job.

2. **Be sneaky, then brand it**
   Her father took the children on outings he wanted and called them special days. Those are the ones they remember.

3. **Seek highs and lows**
   Signature stories are not winning after winning. They need an arc — like James Bond or Jurassic Park.

4. **Disneyland, the worst day**
   Long queues, a dead phone, an au pair with heatstroke. Written up honestly in the holiday card, it became the best story — and people remember the peak and the end.

5 **Bank them, briefly**

Six-word stories ('Married the wrong girl. Fixed it.'). Ask 'what was the story of today?' instead of 'what did you do today?'

WORKED EXAMPLE

When a day goes badly, decide while it is happening that it will be a story: notice the details, take the photo of the unhappy moment, and write the six words that night.

WATCH FOR

Only banking wins. A collection of successes has no arc, and nobody asks to hear it twice.

CHECK YOURSELF · ANSWERS AT THE BACK

3.1 Why carve out an area of incompetence?

A It keeps you humble, which makes you easier to trust
B It generates stories and spares you a task
C It hands responsibility to someone who enjoys it
D It shows the family that experts are boring

3.2 What did her father do with outings the children did not want?

A Cancelled them and promised a better day out later
B Paid the children a little to come along anyway, as a bribe
C Branded them as special days, which made them memorable
D Went alone and told them about it at dinner

3.3 Why was the bad Disneyland day worth recording?

A A signature story needs highs and lows, and people remember the peak and the end
B The tickets were so expensive that the family felt it had to get its money's worth
C Bad days are funnier to retell than good ones, so they are the better stories
D It made the children laugh at the time, which is what a signature story needs

MATH 130 · PROBABILITY · LECTURE 4 OF 14

Expectation, Indicator Random Variables, Linearity

Mon 16 Nov · 50 min video · Watch on YouTube

THE TAKEAWAY

The expected value is a weighted average of values by their probabilities — and LINEARITY, $E(X+Y) = E(X) + E(Y)$ whether or not X and Y are independent, is the single most useful fact about it.

THE ARGUMENT · 17 STEPS

1 **First, what a CDF is for**
Knowing F is knowing the whole distribution: any interval probability comes out of it, since $P(a < X \leq b) = F(b) - F(a)$ by splitting $\{X \leq b\}$ into two disjoint cases.

2 **Three properties, all visible in the picture**
A CDF is increasing (not strictly — flat stretches are allowed), right-continuous (the jumps close on top), and runs to 0 as $x \to -\infty$ and to 1 as $x \to \infty$. And the converse holds: any function with those three is somebody's CDF.

3 **Independence of random variables**
X and Y are independent if the joint CDF factors, $P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y)$, for ALL x and y. In the discrete case the equivalent and far friendlier statement is that the joint PMF factors.

4 **Average, mean, expected value**
Three words for the same thing. Say 'average' with no qualifier and you mean the mean.

5 **Grouped and ungrouped**
To average five 1s, two 3s and one 5, either add all eight and divide by 8, or take the weighted average $1 \cdot \frac{5}{8} + 3 \cdot \frac{2}{8} + 5 \cdot \frac{1}{8}$. Identical. That trivial observation is the whole idea behind the definition that follows — and it is also the proof of linearity.

6 **The definition**
$E(X) = \sum_x x\, P(X = x)$, summed over the values X actually takes. Values weighted by probabilities: likely values count for more.

7 **Bernoulli**
$E(X) = 1 \cdot p + 0 \cdot q = p$. One line, and it is about to carry a great deal of weight.

8 **The fundamental bridge**
Let $I_A = 1$ if A occurs and 0 otherwise — an INDICATOR random variable. Then $E(I_A) = P(A)$. A grandiose name for a one-line identity, but it connects every probability question to an expectation question and lets you cross in either direction.

9 **The binomial mean, the hard way**
$\sum_k k\binom{n}{k}p^k q^{n-k}$, rescued by $k\binom{n}{k} = n\binom{n-1}{k-1}$ — the committee-with-a-president story — then a change of variable, then the binomial theorem. It works. It is unpleasant.

10 **Linearity**
$E(X+Y) = E(X) + E(Y)$, and $E(cX) = cE(X)$. The second is obvious. The first is obvious when X and Y are independent and surprising when they are not — and it is true anyway.

11 **The binomial mean, the easy way**
Bin(n, p) is a sum of n Bernoulli(p)s, each with mean p, so the mean is np. That is a calculation you do in your head, and it never touches the PMF.

12 **The hypergeometric mean, same trick**

Let X_j indicate that the jth card is an ace. Then $X = \sum_{j=1}^{5} X_j$, so $E(X) = 5\,P(\text{first card is an ace}) = 5/13$ — by linearity, then symmetry, then the bridge. The X_j are DEPENDENT and it does not matter.

13 **Which is the real lesson**

Indicators, linearity, symmetry, and the fundamental bridge used together solve problems whose direct PMF sums are hopeless. For the mean, a hypergeometric behaves exactly as if it were binomial.

14 **The geometric distribution**

Independent Bernoulli(p) trials; X counts the FAILURES before the first success. $P(X = k) = q^k p$, because that is the probability of one specific sequence, and only one sequence gives $X = k$. Some books count the success; be careful which convention you are reading.

15 **It is valid because it is a geometric series**

$\sum_{k \geq 0} pq^k = p/(1-q) = p/p = 1$. That is where the name comes from — as with the binomial, the PMF's validity check is the series it is named after.

16 **Its mean, by calculus**

$\sum kq^k$ has an inconvenient k in front. Start from the geometric series, differentiate both sides with respect to q to pull a k down, multiply back by q, and get $E(X) = q/p$.

17 **Its mean, by story**

Let $c = E(X)$. With probability p the first trial succeeds and $X = 0$; with probability q it fails and the problem RESTARTS, so $X = 1+$ the same thing. Hence $c = q(1 + c)$, giving $c = q/p$ with no calculus at all.

WORKED EXAMPLE

When a count is a sum of yes-or-no events, write it as a sum of indicators before doing anything else. Linearity then applies whether or not those events are independent, and the fundamental bridge turns each term into a probability you already know.

WATCH FOR

Believing linearity needs independence. It does not — and the problems where that matters, like counting aces without replacement, are precisely the ones where it saves you.

CHECK YOURSELF · ANSWERS AT THE BACK

4.1 $E(X+Y) = E(X) + E(Y)$ requires:

A X and Y independent
B X and Y identically distributed
C Nothing beyond the two expectations existing
D X and Y non-negative

4.2 The expected number of aces in a five-card hand, to three decimal places, is:

Your answer: _____

4.3 For a geometric distribution counting failures before the first success with $p = 0.2$, the mean, to three decimal places, is:

Your answer: _____

SPCH 100 · STORYTELLING · LECTURE 5 OF 14

Twelve lessons: zoom in, quote it, use 'how are you?'

Mon 16 Nov · 12 min video · Watch on YouTube

THE TAKEAWAY

Philipp Humm's twelve lessons fall into two groups: how to get more stories (say yes to fear, find magic in the mundane, use 'how are you?' as practice, keep a bank) and how to tell them (start near the challenge, zoom in, quote the dialogue and your inner voice, give the why, show the change).

THE ARGUMENT · 6 STEPS

1. **Share the rock bottom**
   Dan Martell skipped his teenage years of drugs and crime because they made him emotional. Success impresses people; struggle changes them.

2. **Start at the challenge**
   A workshop participant spent 93 seconds on job title and background before his story began, and lost the room.

3. **Zoom in**
   Not the helicopter shot ('I faced a big challenge') but the trench: where am I, what am I doing, thinking, feeling, hearing?

4. **Quote it**
   Outer dialogue ('Phillip, what the hell was that?') and inner dialogue ('how am I going to turn this around?') make the story play like a film.

5 **The why and the change**

Beginners give events, amateurs give the goal, pros give the reason it mattered. Memorable stories show a moment of change.

6 **Get more stories**

Say yes to what scares you; find magic in the mundane with Homework for Life; answer 'how are you?' with a tiny true story; keep a story bank.

WORKED EXAMPLE

Before telling a moment, close your eyes and answer the five zoom questions aloud. Use one sensory detail, one quoted line, and one line of inner voice. Cut the background.

WATCH FOR

Staying in the helicopter: 'it was a stressful period at work'. Nothing is visible, so nothing is felt.

CHECK YOURSELF · ANSWERS AT THE BACK

5.1 What does 'zooming in' mean in Humm's war-film image?

A Describing the whole battlefield before showing any one soldier
B Dropping into one specific scene with its sights, sounds and thoughts
C Cutting quickly between scenes so the story moves at the pace of a film
D Telling the story from the point of view of someone watching from afar

5.2 According to Humm, what separates a pro storyteller from an amateur?

A Starting with background, so the audience knows who you are
B Sharing what the goal was clearly, before saying how it went
C Sharing why the goal mattered, not just what the goal was
D Ending with a lesson stated plainly, so nobody can miss it

5.3 What does he suggest doing when someone asks 'how are you?'

A Tell a tiny, true story from your day
B Say 'busy' and ask how they are instead
C Ask them first, then match their answer
D Change the subject to something funnier

SPCH 100 · STORYTELLING · LECTURE 6 OF 14

Most of the work is finding — and killing

Mon 16 Nov · 4 min video · Watch on YouTube

THE TAKEAWAY

Ira Glass on the part nobody tells you: finding a decent story takes as long as producing it, a good team kills a third to a half of what it tries, and luck comes from putting enough material through your hands.

THE ARGUMENT · 5 STEPS

1. **Finding is the work**
   We imagine the real work is shooting and editing; often finding the decent story takes longer than making it.

2. **Half the week is looking**
   His staff do nothing but look for stories and try them, and kill between a third and a half of everything they try.

3. **Enjoy the killing**
   When the feeling you had is not in the tape, kill it. Killing it makes room for something better.

4. **Everything is trying to be crap**
   Like entropy: anything you record drifts toward pointless, digressive and boring unless you prop it up aggressively at every stage.

5. **Get lucky on a schedule**
   Do enough interviews every week that once a month you stumble on something so good it pays for the other weeks.

WORKED EXAMPLE

Treat your story bank as a pipeline, not a vault: try stories out loud on friends, keep the ones that land, and cut the rest without arguing with yourself.

WATCH FOR

Keeping a story because you loved the moment, even though it falls flat every time you tell it. The feeling you had is not automatically in the telling.

CHECK YOURSELF · ANSWERS AT THE BACK

6.1 Roughly what share of the stories his team tries end up killed?

A Almost none, once a story is chosen
B Between a third and a half
C About nine in every ten tried
D Exactly one in every ten

6.2 What does he mean by 'all production is trying to be crap'?

A Equipment fails so often that most recordings can't be used in the end at all
B Interviewees perform for the microphone, so most of what they say is fake
C Material drifts toward pointless and boring unless you fight it at every stage
D Audiences are so hard to please that most finished stories will disappoint

6.3 How do you put yourself in a position to get lucky?

A Do enough material on a schedule that a great story turns up every month
B Wait for a story that excites you, and only then commit time to producing it
C Only chase big stories, since small ones rarely become great ones in the end
D Study stories that worked before and copy their structure as closely as you can

MATH 110 · LINEAR ALGEBRA · LECTURE 7 OF 14

Solving Ax = 0: Pivot Variables, Special Solutions

Tue 17 Nov · 43 min video · Watch on YouTube

THE TAKEAWAY

Elimination does not change the null space, and what it leaves behind — the pivot columns — tells you exactly how many free choices the solutions have.

THE ARGUMENT · 6 STEPS

1. **Elimination preserves the null space, not the column space**
   Subtracting a multiple of one equation from another leaves the solution set alone, so $A\vec{x} = \vec{0}$ and $U\vec{x} = \vec{0}$ have the same answers. The columns themselves do move, so $C(U)$ is a different space from $C(A)$.

2. **A zero in the pivot position with zeros beneath is not a failure**
   In the square case it meant singular and you stopped. Rectangular, you shrug and move to the next column. What comes out is echelon form — a staircase, not a triangle — and the count of pivots is the rank r.

3. **Pivot columns and free columns**
   A column with a pivot has a pivot variable; a column without one has a free variable. Free means exactly what it says: assign it any value at all, and back substitution then fixes the pivot variables.

4. **One special solution per free variable**
   Set one free variable to 1 and the rest to 0, solve, and repeat for each. That gives $n - r$ vectors, and every vector in the null space is a combination of them. The count is the whole content of the algorithm.

5. **Reduced row echelon form makes the answer readable**
   Eliminate upwards as well and scale each pivot to 1. Now R carries an identity in the pivot rows and columns, a free block F beside it, and zero rows below — one for each row that was a combination of the others.

6. **The special solutions are $-F$ stacked on I**
   $R = \begin{pmatrix} I & F \end{pmatrix}$ and $R\vec{x} = \vec{0}$ reads $\vec{x}_{pivot} = -F\vec{x}_{free}$. Put the identity in the free rows and $-F$ appears above it. The back substitution has already been done; the sign flip is the only work.

WORKED EXAMPLE

$A = \begin{pmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{pmatrix}$. Column two is twice column one and row three is row one plus row two, so elimination finds two pivots and a zero row. Four columns, rank two: two free variables, two special solutions, and the null space is a plane.

WATCH FOR

Reading a row of zeros as a mistake in the arithmetic. It is a finding: that row was a combination of the others, and elimination is reporting it.

KEY CONCEPTS

Rank plus nullity is conserved

CHECK YOURSELF · ANSWERS AT THE BACK

7.1 Elimination leaves the null space unchanged but changes the column space because:

A It changes the number of pivots
B Row operations preserve the solutions of $A\vec{x} = \vec{0}$ while moving the columns themselves
C The null space is always zero
D Only column operations matter

7.2 For $A = \begin{pmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{pmatrix}$, the number of special solutions is:

Your answer: _____

SPCH 100 · STORYTELLING · LECTURE 8 OF 14

The taste gap — volume on a deadline

Tue 17 Nov · 5 min video · Watch on YouTube

THE TAKEAWAY

Ira Glass's best-known advice: for years your taste will be better than your work, which is why most people quit — and the only way to close the gap is a large volume of work on a deadline.

THE ARGUMENT · 5 STEPS

1. **Taste got you in**
   You want to make things because you love them, so your taste is good from the start.

2. **The gap**
   For the first couple of years what you make is not that good, and your taste is good enough to tell. Many people quit there.

3 **Everyone goes through it**
Nearly everyone he knows who does interesting creative work spent years in that phase. It is normal.

4 **Volume on a deadline**
Finish one story every week or month, ideally with someone expecting it. Only volume closes the gap.

5 **Year eight**
He plays a clip of himself eight years in: ill-conceived writing, every third word stressed. Talk the way people normally talk.

WORKED EXAMPLE
Set a weekly deadline with an external witness — someone who expects the recording on Friday — and keep it even when the piece disappoints you.

WATCH FOR
Waiting until the work matches your taste before you put it out. The work only catches up through the volume you are postponing.

CHECK YOURSELF · ANSWERS AT THE BACK

8.1 What is 'the gap' Glass describes?

A The time it takes to go from one finished project to the next
B The distance between your good taste and your not-yet-good work
C The distance between what your audience wants and what you make
D The pause a skilled storyteller leaves before the key line lands

8.2 What does he say closes the gap?

A Better equipment, once you can afford it
B Waiting until inspiration finally arrives
C Studying the theory of story before writing
D A large volume of work on a deadline

8.3 What delivery fault does he point out in his own year-eight tape?

A Stressing every third word instead of talking the way people talk
B Speaking so fast that the listener couldn't follow the story at all
C Pausing so often that the story lost all of its forward momentum
D Reading from notes, so his voice sounded flat and rehearsed throughout

SPCH 100 · STORYTELLING · LECTURE 9 OF 14

The voice toolbox — register, prosody, pace, and a warm-up

Tue 17 Nov · 10 min video · Watch on YouTube

THE TAKEAWAY

Julian Treasure's TED talk in three parts: seven habits that make people stop listening, four foundations that spell HAIL, and a toolbox of register, timbre, prosody, pace, pitch and volume — warmed up before anything important.

THE ARGUMENT · 5 STEPS

1. **Seven deadly sins**
   Gossip, judging, negativity, complaining, excuses, exaggeration that turns into lying, and dogmatism.

2. **HAIL**
   Honesty, authenticity, integrity, love. Love tempers honesty, and it is hard to judge someone while wishing them well.

3. **Register and timbre**
   Nose, throat or chest: depth carries authority. Timbre — rich, smooth, warm — can be trained with breathing, posture and exercises.

4. **Prosody, pace, pitch, volume**
   Avoid one note and avoid making every statement sound like a question. Use silence. Get quiet to draw people in.

5. **The warm-up**
   Arms up and sigh, lip buzz, tongue la-la, rolled R, and the siren from high to low.

WORKED EXAMPLE

Before an interview, do the warm-up in the corridor (the siren if you only have time for one), then say your opening line once from the chest, slowly, with silence before the key word.

WATCH FOR

Upspeak: making every sentence end like a question. It takes away a prosody tool you need for real questions and makes statements sound unsure.

CHECK YOURSELF · ANSWERS AT THE BACK

9.1 What does HAIL stand for?

A Humour, anecdote, insight, and a lesson
B Honesty, authenticity, integrity, love
C Hear, acknowledge, inquire, then lead
D Honesty, attention, intention, listening

9.2 Where does Treasure say to speak from when you want weight?

A The nose
B The throat
C The chest
D The head voice

9.3 Which warm-up does he say he would keep if he could only do one?

A The siren, from high 'we' to low 'aw'
B The lip buzz, held on one long breath
C The rolled R, to wake up the tongue
D The long sigh, from high to low

MATH 120 · SINGLE-VARIABLE CALCULUS · LECTURE 10 OF 14

Exponential and Log

Wed 18 Nov · 48 min video · Watch on YouTube

THE TAKEAWAY

There is exactly one base whose exponential is its own derivative, and defining e to be that base is what makes every other exponential and every logarithm differentiable.

THE ARGUMENT · 8 STEPS

1. **The difference quotient factors**
   For $f(x) = a^x$, $\dfrac{a^{x+\Delta x} - a^x}{\Delta x} = a^x \cdot \dfrac{a^{\Delta x} - 1}{\Delta x}$. The a^x comes out because the exponential law says so, and what is left does not mention x at all.

2. **So the whole problem is one number**
   That leftover limit is a constant depending only on a. Call it $M(a)$. Then $\frac{d}{dx}a^x = M(a)\,a^x$ — every exponential is proportional to its own derivative, and $M(a)$ is the constant of proportionality.

3 **And that number is a slope you can point at**

Put $x = 0$: $\frac{d}{dx}a^x\big|_0 = M(a) \cdot a^0 = M(a)$. So $M(a)$ is simply the slope of $y = a^x$ where it crosses the y-axis at height 1.

4 **Define e as the base that makes it 1**

$M(2) \approx 0.693$ and $M(4) \approx 1.386$ — under 1 and over 1 — so somewhere between is a base with $M = 1$. Name it e. Then $\frac{d}{dx}e^x = e^x$, exactly.

5 **Why such a base has to exist**

Replacing a by a^k stretches the graph horizontally by $1/k$, which multiplies the slope at 0 by k : $M(a^k) = kM(a)$. Choosing $k = 1/M(2)$ manufactures the base you want, so $e = 2^{1/M(2)}$.

6 **The log comes back by implicit differentiation**

$y = \ln x$ means $e^y = x$. Differentiate: $e^y y' = 1$, so $y' = 1/e^y = 1/x$. The awkward function $\ln$ has the cleanest derivative in the subject.

7 **Two ways to differentiate any a^x**

Either rewrite $a^x = e^{(\ln a)x}$ and use the chain rule, or take logs first — $\ln u = x \ln a$, differentiate both sides to get $u'/u = \ln a$. Both give $\frac{d}{dx}a^x = (\ln a)a^x$, and both show $M(a) = \ln a$.

8 **Logarithmic differentiation earns its keep on x^x**

It is neither a power (x^n, fixed exponent) nor an exponential (a^x, fixed base), so no rule applies. But $\ln u = x \ln x$, so $u'/u = 1 + \ln x$, and $\frac{d}{dx}x^x = x^x(1 + \ln x)$.

WORKED EXAMPLE

The compound-interest limit $\lim_{n\to\infty}(1 + 1/n)^n$ is the derivative of $\ln$ at $x = 1$ in disguise. Since $\ln'(1) = 1$, the difference quotient $\frac{\ln(1+1/n) - \ln 1}{1/n} = n\ln(1 + 1/n) = \ln\left((1 + 1/n)^n\right)$ tends to 1 — so the thing inside the log tends to e.

WATCH FOR

Treating $M(a)$ as if it were 1 for every base. It is 1 only for e; in general it is $\ln a$, which is why $\frac{d}{dx}2^x$ is $0.693 \cdot 2^x$ and not 2^x.

CHECK YOURSELF · ANSWERS AT THE BACK

10.1 What is $M(a)$, the constant with $\frac{d}{dx}a^x = M(a)a^x$?

A The value of a itself

B The slope of $y = a^x$ at $x = 0$, which turns out to be $\ln a$

C Always 1

D The y-intercept of a^x

10.2 The derivative of x^x at $x = 1$ is:

Your answer: _____

SPCH 100 · STORYTELLING · LECTURE 11 OF 14

The five dials — rate, volume, melody, tone, pause

Wed 18 Nov – Thu 19 Nov · day 1–2 of 3 · 43 min video · Watch on YouTube · finishes in week 6

THE TAKEAWAY

Vinh Giang's five foundations — rate, volume, pitch as melody, tonality, and the pause — coached live on a student, then applied to accents, interviews, imitation and nerves. The voice is behaviour, so it can change.

THE ARGUMENT · 6 STEPS

1. **You are not trapped in your voice**
   Changing your mouth movements and breath changes how you sound. The way you speak is a set of behaviours.

2. **The five dials**
   Vary the rate. Speak at a five, not a three. Use melody, because a song sticks and a page does not. Let your face carry the emotion. Pause.

3. **Live coaching**
   A student answers 3-2-1 style, then again with more volume, gestures and a lower pitch at the end of sentences, then with long pauses and no opening 'so'.

4. **Accent**
   Record twenty improvised minutes, send it to a speech pathologist for pronunciation and to an ESL teacher for grammar, then fix one thing at a time.

5. **Interviews and imitation**
   Spend 5–10 per cent of the time on connection: answer 'tell me about yourself' with a prepared story. Copy great speakers the way a chef copies recipes, then add your own lemon.

6. **Nerves and range**
   Burn off adrenaline with a quick walk or push-ups, use breathing, and focus on the audience. When an unused part of your voice feels fake, tell yourself it is only unfamiliar.

WORKED EXAMPLE

His interview answer: a prepared story (selling MP3 players at 13, and what it taught him about selling the wrong way and the right way), linked to the job, takes about three minutes and builds rapport before the questions start.

WATCH FOR

Calling an unfamiliar voice 'fake'. It is the same instrument — you have just never used those keys, and calling them fake stops you exploring.

CHECK YOURSELF · ANSWERS AT THE BACK

11.1 Why does Giang compare a song with a page of a book?

A Songs are shorter, so the listener can hold the whole thing in their head
B Both hold about as many words, but the melody makes the song memorable
C A song repeats its chorus, so repetition is what makes it stick
D Reading a page aloud is harder than singing, so it sounds flat

11.2 What does he say causes a monotone voice?

A Speaking too fast to let the pitch move
B Too much volume, which flattens the melody
C A monotone body — including a neutral face
D Reading from notes instead of talking

11.3 What is his advice for someone worried about their accent?

A Have a speech pathologist and an ESL teacher review twenty improvised minutes
B Copy one native speaker's accent exactly, sound by sound, until it is yours
C Slow right down and fix every sound at once, so the accent fades in weeks
D Keep quiet in meetings until a coach says your accent has become neutral

MATH 130 · PROBABILITY · LECTURE 12 OF 14

Expectation Continued

Thu 19 Nov · 50 min video · Watch on YouTube

THE TAKEAWAY

Linearity is proved by summing over the SAMPLE SPACE rather than over values — and once you have it, indicators plus symmetry crack problems that defeat the direct approach entirely.

THE ARGUMENT · 14 STEPS

1 **Why the direct proof gets stuck**
Write $E(X+Y)$ from the definition and you are summing over values of the sum; write $E(X)+E(Y)$ and you are summing over values of X and of Y separately. There is no obvious way to turn one into the other, and conditioning on X leaves you with a term you cannot simplify unless X and Y are independent.

2 **Sum over pebbles instead**
$E(X)=\sum_{s\in S} X(s)P(\{s\})$ — average over the individual outcomes, ungrouped, rather than over the distinct values. This is the same ungrouped-versus-grouped observation as before, and it is the same number.

3 **And then the proof is two lines**
$E(X+Y) = \sum_s (X(s)+Y(s))P(s) = \sum_s X(s)P(s) + \sum_s Y(s)P(s) = E(X)+E(Y)$. Everything is summed over the SAME index, so the sum just splits. Nothing anywhere asked whether X and Y are related.

4 **Check it at the extremes**
If $X = Y$ — maximal dependence — then $E(X+Y) = E(2X) = 2E(X) = E(X)+E(Y)$. If they are independent it is intuitive. The proof covers everything in between.

5 **The negative binomial**
Not a negated binomial. It generalises the geometric: independent Bernoulli(p) trials, count the failures before the rth success. $P(X=n) = \binom{n+r-1}{r-1} p^r q^n$ — the last trial must be a success, and the earlier $n+r-1$ trials are any arrangement of $r-1$ successes among them.

6 **Its mean, in your head**
Wait for the first success, then the additional wait for the second, and so on: $X = X_1 + \cdots + X_r$ with each X_j geometric(p). By linearity $E(X) = rq/p$. Summing n against that PMF gets the same answer and takes a page.

7 **The first success distribution**
$\mathrm{FS}(p)$ counts the trials INCLUDING the success, so $Y = X + 1$ with X geometric. Then $E(Y) = q/p + 1 = 1/p$ — if you succeed one time in ten, it takes about ten tries. Conventions differ between textbooks, so convert deliberately rather than copying a formula.

8 **A Putnam problem**
Permute $1, \ldots, n$ at random. A LOCAL MAXIMUM is an entry larger than its neighbours — for the two endpoints, larger than its single neighbour. Find the expected number of them. Competitors that year rated it among the hardest on the exam.

9 **Indicators, then two cases**
Let I_j indicate a local maximum at position j, so the count is $\sum_j I_j$. Interior: among the three entries at $j-1, j, j+1$, the largest is equally likely to be in any of the three positions, so $P = 1/3$. Endpoint: only one neighbour, so $P = 1/2$.

10 **Which makes it a one-line answer**
$\frac{n-2}{3} + 2 \cdot \frac{1}{2} = \frac{n+1}{3}$. Check $n = 2$: the answer is 1, and indeed both 12 and 21 have exactly one local maximum.

11 **And the tempting wrong answer**
Saying $1/4$ for an interior position — a half for beating the left neighbour times a half for beating the right. Those two events are not independent, exactly as 'A older than B' and 'A older than C' are not.

12 **The St Petersburg paradox**
Flip a fair coin until heads; you are paid 2^X where X counts flips including the head. $E(2^X) = \sum_{k \geq 1} 2^k \cdot 2^{-k} = 1 + 1 + 1 + \cdots = \infty$. So the fair price to play is unbounded, and almost nobody will offer more than a few tens of dollars.

13 **The resolution is that nobody has infinite money**

Cap the payout at 2^{40}, over a trillion dollars, and the sum has forty terms: the game is worth \$40, or \$41 if the house pays the cap rather than defaulting. An astronomically generous cap still leaves a very cheap game, which is not paradoxical at all.

14 **And the warning it contains**

$E(2^X) \neq 2^{E(X)}$ — here ∞ against $2^2 = 4$. Expectation moves through sums and constants, and through nothing else. Sliding an E past a nonlinear function is the most expensive habit in the subject.

WORKED EXAMPLE

For 'expected number of things with a property', define one indicator per candidate, add them, and use linearity. Then compute a single one of those probabilities by symmetry — and check whether the boundary cases need their own count.

WATCH FOR

Moving the expectation inside a nonlinear function. $E(g(X))$ and $g(E(X))$ are different numbers, and St Petersburg shows how far apart they can be.

CHECK YOURSELF · ANSWERS AT THE BACK

12.1 The expected number of local maxima in a random permutation of $1, \ldots, 7$, to three decimal places, is:

Your answer: _____

12.2 The mean of a negative binomial with $r = 3$ and $p = 0.4$, counting failures before the third success, to three decimal places, is:

Your answer: _____

12.3 In the St Petersburg game the expected payout is infinite, yet capping it at 2^{40} dollars makes the game worth about forty dollars. What does this show?

A The expectation was computed wrongly
B That an infinite expectation can be extremely sensitive to a bound that is itself astronomically large
C That expectation is the wrong concept for gambling
D That the coin cannot really be fair

AI 200 · NEURAL NETWORKS: ZERO TO HERO · LECTURE 13 OF 14

Intro to Large Language Models

Thu 19 Nov – Fri 20 Nov · day 1–2 of 2 · 60 min video · Watch on YouTube

NOT YET SUMMARISED

This lecture has no summary yet, so nothing is printed for it. It is listed with its video link on “Not yet summarised” at the back.

MATH 110 · LINEAR ALGEBRA · LECTURE 14 OF 14

Solving Ax = b: Row Reduced Form R

Fri 20 Nov · 47 min video · Watch on YouTube

THE TAKEAWAY

Every solution of $A\vec{x} = \vec{b}$ is one particular solution plus anything from the null space, and the rank alone decides whether there are none, one, or infinitely many.

THE ARGUMENT · 6 STEPS

1. **Carry $\vec{b}$ through elimination as an extra column**
   Work on the augmented matrix $[A;|;\vec{b}]$ so the right-hand side gets every operation the left does. If a combination of rows gives a zero row, the same combination of b's entries must give zero, or the system is inconsistent.

2. **That is solvability, stated twice**
   "$\vec{b}$ lies in the column space" and "every combination of rows that zeroes A also zeroes $\vec{b}$" are the same condition from opposite sides. Elimination reports the second; the column picture explains the first.

3. **The particular solution: set every free variable to zero**
   That leaves a square system in the pivot variables, which back substitution solves. It is one solution among many, chosen because it is the cheapest one to write down.

4. **Add the null space to get the rest**
   $A\vec{x}_p = \vec{b}$ and $A\vec{x}_n = \vec{0}$, so $A(\vec{x}_p + \vec{x}_n) = \vec{b}$ for every $\vec{x}_n$. The complete solution is $\vec{x}_p$ plus all combinations of the special solutions — and this pattern recurs everywhere linear equations do.

5. **The solution set is a shifted subspace**
   The null space is a subspace through the origin; the solution set is that same flat object translated to sit on $\vec{x}_p$. It does not contain $\vec{0}$, so it is not a subspace — which is why no constant may multiply $\vec{x}_p$.

6 Rank decides the count before any arithmetic

$r = m = n$: invertible, exactly one. $r = n < m$: no free variables, so zero or one. $r = m < n$: no zero rows, so always at least one, hence infinitely many. $r < m$ and $r < n$: zero or infinitely many.

WORKED EXAMPLE

$A = \begin{pmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{pmatrix}$ with $\vec{b} = (1, 5, 6)$: row three is row one plus row two and $1 + 5 = 6$, so it is solvable. Set $x_2 = x_4 = 0$, leaving $x_1 + 2x_3 = 1$ and $2x_3 = 3$, so $\vec{x}_p = (-2, 0, \frac{3}{2}, 0)$.

WATCH FOR

Putting a free constant in front of $\vec{x}_p$. Scaling it breaks $A\vec{x}_p = \vec{b}$; only the null space part is free to be multiplied and added.

KEY CONCEPTS

Rank plus nullity is conserved

CHECK YOURSELF · ANSWERS AT THE BACK

14.1 The set of solutions to $A\vec{x} = \vec{b}$ with $\vec{b} \neq \vec{0}$ is:

A A subspace
B The null space shifted so it passes through one particular solution
C Always a single point
D Always empty

14.2 For that A and $\vec{b} = (1, 5, 6)$, the first component of the particular solution with free variables set to zero is:

Your answer: _____

WEEK 5 · SUN 15 – FRI 20 NOVEMBER 2026

Not yet summarised

1 lecture this week has no summary yet. Watch or rewatch them from the links below.

LECTURES

13 Intro to Large Language Models

AI 200 · Thu 19 Nov – Fri 20 Nov · day 1–2 of 2 · 60 min video

https://www.youtube.com/watch?v=zjkBMFhNj_g

WEEK 5 · SUN 15 – FRI 20 NOVEMBER 2026

What this week established

Each lecture’s takeaway, by course.

MATH 120 · Single-Variable Calculus

1 **Implicit Differentiation.** Differentiate the equation you were given instead of solving it first, and both fractional powers and every inverse function fall out.

10 **Exponential and Log.** There is exactly one base whose exponential is its own derivative, and defining e to be that base is what makes every other exponential and every logarithm differentiable.

AI 200 · Neural Networks: Zero to Hero

2 **Let's reproduce GPT-2 (124M).** Reproducing GPT-2 (124M) is the GPT from before plus everything that makes it real: the exact architecture so OpenAI's weights load, GPT-2's initialisation, a long list of speed-ups, GPT-3's optimisation recipe — AdamW, clipping, warmup and cosine decay, a 0.5M-token batch via gradient accumulation — and honest evaluation on held-out loss and HellaSwag.

13 **Intro to Large Language Models.** Not yet summarised.

SPCH 100 · Storytelling

3 **Personal signature stories — make them, then bank them.** Aaker's family rules for creating signature stories, not just finding them: carve out areas of incompetence, be sneaky and brand ordinary days as special, seek highs and lows, and bank the results — six words is enough.

5 **Twelve lessons: zoom in, quote it, use 'how are you?'.** Philipp Humm's twelve lessons fall into two groups: how to get more stories (say yes to fear, find magic in the mundane, use 'how are you?' as practice, keep a bank) and how to tell them (start near the challenge, zoom in, quote the dialogue and your inner voice, give the why, show the change).

6 **Most of the work is finding — and killing.** Ira Glass on the part nobody tells you: finding a decent story takes as long as producing it, a good team kills a third to a half of what it tries, and luck comes from putting enough material through your hands.

8 **The taste gap — volume on a deadline.** Ira Glass's best-known advice: for years your taste will be better than your work, which is why most people quit — and the only way to close the gap is a large volume of work on a deadline.

9 **The voice toolbox — register, prosody, pace, and a warm-up.** Julian Treasure's TED talk in three parts: seven habits that make people stop listening, four foundations that spell HAIL, and a toolbox of register, timbre, prosody, pace, pitch and volume — warmed up before anything important.

11 **The five dials — rate, volume, melody, tone, pause.** Vinh Giang's five foundations — rate, volume, pitch as melody, tonality, and the pause — coached live on a student, then applied to accents, interviews, imitation and nerves. The voice is behaviour, so it can change.

MATH 130 · Probability

4 **Expectation, Indicator Random Variables, Linearity.** The expected value is a weighted average of values by their probabilities — and LINEARITY, $E(X+Y) = E(X) + E(Y)$ whether or not X and Y are independent, is the single most useful fact about it.

12 **Expectation Continued.** Linearity is proved by summing over the SAMPLE SPACE rather than over values — and once you have it, indicators plus symmetry crack problems that defeat the direct approach entirely.

MATH 110 · Linear Algebra

7 **Solving Ax = 0: Pivot Variables, Special Solutions.** Elimination does not change the null space, and what it leaves behind — the pivot columns — tells you exactly how many free choices the solutions have.

14 **Solving Ax = b: Row Reduced Form R.** Every solution of $A\vec{x} = \vec{b}$ is one particular solution plus anything from the null space, and the rank alone decides whether there are none, one, or infinitely many.

WEEK 5 · SUN 15 – FRI 20 NOVEMBER 2026

Answers

The answers and explanations are the summaries' own.

1 Implicit Differentiation

1.1 B — You never have to differentiate the quartic formula — two lines give y' in terms of x and y

The explicit solution exists but is a nest of radicals with several branches. Implicit differentiation needs a point on the curve, but the differentiation itself stays trivial.

1.2 0.2

$\frac{d}{dx}\arctan x = \frac{1}{1+x^2}$, so at $x = 2$ it is $\frac{1}{5} = 0.2$. The arctangent is flattening towards its horizontal asymptote at $\pi/2$.

2 Let's reproduce GPT-2 (124M)

2.1 38597376

Each of the two matrices is 50,257 × 768 = 38,597,376 parameters — about 30% of the 124M model. The notebook shows OpenAI's checkpoint stores lm_head and wte with the same values at the same memory address; in the code it is one line, wte.weight = lm_head.weight.

2.2 B — Each block adds two outputs to the residual stream; this stops its std growing

Every block adds attention's and the MLP's projections into the stream, 2 · n_layer additions in all, and adding n random contributions grows the std like $\sqrt{n}$. The notebook adds 100 random vectors scaled by $100^{-1/2}$ and the std stays about 1. Hence the c_proj layers, the ones writing into the stream, are scaled down.

2.3 8

One micro-step processes 64 × 1,024 = 65,536 tokens; 524,288 / 65,536 = 8. With DDP on 8 GPUs the divisor gains a factor of 8 and it becomes 1 micro-step per GPU. The total is 2^19, about 0.5M tokens — GPT-3's batch size for this model size.

3 Personal signature stories — make them, then bank them

3.1 B — It generates stories and spares you a task

Her father's photography and her husband's bins are both stories and both chores avoided.

3.2 C — Branded them as special days, which made them memorable

Anticipating something as special made it remembered as special.

3.3 A — A signature story needs highs and lows, and people remember the peak and the end

Designing for memories rather than experiences is what creates a life of stories.

4 Expectation, Indicator Random Variables, Linearity

4.1 C — Nothing beyond the two expectations existing

Linearity holds for dependent random variables too. The proof sums over the sample space rather than over values, which never asks how X and Y relate.

4.2 0.385

$5/13 = 0.3846$. Five indicators, each with probability $4/52$, added by linearity — the dependence between them changes nothing.

4.3 4

$q/p = 0.8/0.2 = 4$. With a one-in-five chance each trial you expect four failures before the first success, and five trials including it.

5 Twelve lessons: zoom in, quote it, use 'how are you?'

5.1 B — Dropping into one specific scene with its sights, sounds and thoughts

Ask where am I, what am I doing, thinking, feeling and hearing.

5.2 C — Sharing why the goal mattered, not just what the goal was

The what gives information; the why gives emotion.

5.3 A — Tell a tiny, true story from your day

Make your life the arena: every small exchange is practice.

6 Most of the work is finding — and killing

6.1 B — Between a third and a half

And he describes the team as among the best at the job.

6.2 C — Material drifts toward pointless and boring unless you fight it at every stage

He compares it to entropy.

6.3 A — Do enough material on a schedule that a great story turns up every month

Volume on a schedule is what produces the rare great one.

7 Solving Ax = 0: Pivot Variables, Special Solutions

7.1 B — Row operations preserve the solutions of $A\vec{x} = \vec{0}$ while moving the columns themselves

Subtracting one equation from another cannot gain or lose a solution. But the columns of U are different vectors from the columns of A, so their combinations span a different space.

7.2 2

Four columns and rank two, so $n - r = 4 - 2 = 2$ free variables — one special solution each.

8 The taste gap — volume on a deadline

8.1 B — The distance between your good taste and your not-yet-good work

You can see your work disappoints you precisely because your taste is good.

8.2 D — A large volume of work on a deadline

Ideally with someone waiting for the work, even if they do not pay you.

8.3 A — Stressing every third word instead of talking the way people talk

Underlining every third word for emphasis sounds unnatural.

9 The voice toolbox — register, prosody, pace, and a warm-up

9.1 B — Honesty, authenticity, integrity, love

Four foundations to stand on so that what you say is welcomed.

9.2 C — The chest

He links depth to perceived power and authority.

9.3 A — The siren, from high 'we' to low 'aw'

He says the pros call it the siren.

10 Exponential and Log

10.1 B — The slope of $y = a^x$ at $x = 0$, which turns out to be $\ln a$

Setting $x = 0$ in the formula gives $M(a)$ directly as the slope at the y-intercept, and the base-e rewrite identifies it as $\ln a$.

10.2 1

$\frac{d}{dx}x^x = x^x(1 + \ln x)$, and at $x = 1$ that is $1 \cdot (1 + 0) = 1$. Neither the power rule nor the exponential rule applies here — logarithmic differentiation is what gets it.

11 The five dials — rate, volume, melody, tone, pause

11.1 B — Both hold about as many words, but the melody makes the song memorable

So speaking with more melody makes what you say stick.

11.2 C — A monotone body — including a neutral face

Your face controls the emotion under your words.

11.3 A — Have a speech pathologist and an ESL teacher review twenty improvised minutes

Then work on one thing at a time, so you do not freeze up.

12 Expectation Continued

12.1 2.667

$(n + 1)/3 = 8/3 = 2.667$. Five interior positions at $1/3$ each plus two endpoints at $1/2$ each.

12.2 4.5

$rq/p = 3(0.6)/0.4 = 4.5$. Three independent waits, each a geometric with mean $q/p = 1.5$.

12.3 B — That an infinite expectation can be extremely sensitive to a bound that is itself astronomically large

Each of the forty surviving terms contributes exactly one dollar, and everything beyond a trillion dollars contributes at most one more. The infinity was carried entirely by outcomes nobody could ever be paid for.

14 Solving Ax = b: Row Reduced Form R

14.1 B — The null space shifted so it passes through one particular solution

It is the null space translated by $\vec{x}_p$ — the same shape and the same dimension, but not through the origin, so it fails the definition of a subspace.

14.2 −2

$2x_3 = 3$ gives $x_3 = \frac{3}{2}$, and then $x_1 = 1 - 2 \cdot \frac{3}{2} = -2$.